

read wikipedia for these r.v.s

$$\text{CDF: } F_X(x_1) = P_X((-\infty, x_1])$$

↪ non-decreasing, right continuous

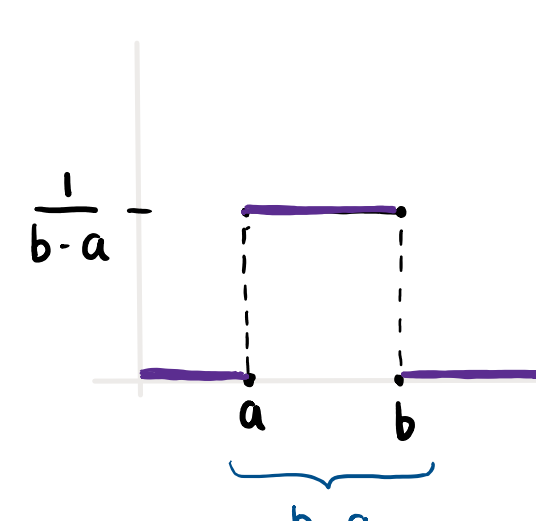
Continuous r.v.s have a continuous $F_X(\cdot)$.

They also have a pdf $f_X(\cdot)$.

$$P_X(a \leq X \leq b) = \int_a^b f_X(u) \cdot du \quad (\text{area under the curve})$$

$$= F_X(b) - F_X(a)$$

- Uniform random variable ($U[a, b]$).
= takes values uniformly at random in $[a, b]$.

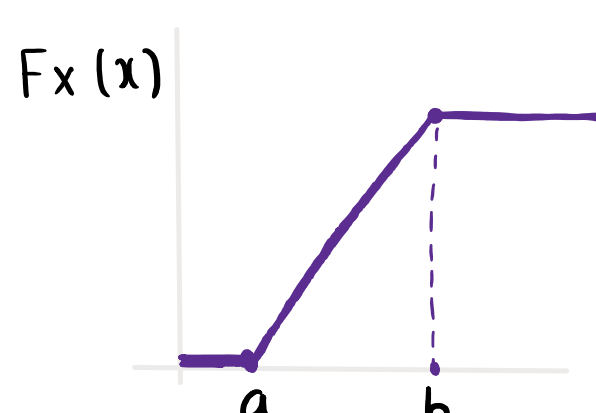


$$\text{area} = 1 \Rightarrow \text{pdf} = \frac{1}{b-a}$$

$$(b-a) \times x = 1$$

$$x = \frac{1}{b-a}$$

$$\text{CDF: } F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & x > b \end{cases}$$



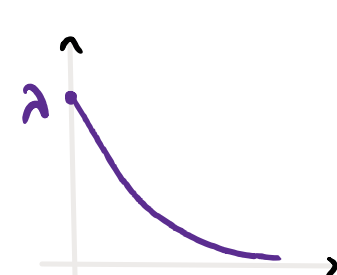
$$E(X) = \frac{a+b}{2}$$

midpoint

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

- Exponential random variable ($\text{Exp}(\lambda)$)
= non-negative r.v. with parameter λ .

$$\text{PDF: } f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$



$$F_X(a) = \int_{-\infty}^a f_X(u) \cdot du = \int_0^a f_X(u) \cdot du$$

$$= \int_0^a \lambda e^{-\lambda u} du = \lambda \left[\frac{e^{-\lambda u}}{-\lambda} \right]_0^a = \lambda \left(\frac{e^{-\lambda a}}{-\lambda} - \frac{1}{-\lambda} \right) = 1 - e^{-\lambda a}$$

$\therefore$ TAIL CDF = $e^{-\lambda a}$

$$\text{Mean} = \int_0^{\infty} u \cdot \lambda \cdot e^{-\lambda u} \cdot du = \lambda \left[u \cdot \int e^{-\lambda u} du - \int \frac{du}{du} \cdot \int e^{-\lambda u} \cdot du \cdot du \right]_0^{\infty} = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

$$E(X^n) = \frac{n!}{\lambda^n}$$

$$* \text{ Integral of tail CDF} = \int_0^{\infty} F_X(x) \cdot dx = \frac{1}{\lambda} = \text{MEAN}$$

IS THIS TRUE IN GENERAL?

PROVE IN GENERAL.

express moment using tail cdf?

$$\text{MEMORYLESS PROPERTY: } P(X > a+t | X > t) = \frac{e^{-\lambda(a+t)}}{e^{-\lambda t}} = e^{-\lambda a}$$

- GAUSSIAN RANDOM VARIABLE ($N(\mu, \sigma^2)$)
= real valued r.v.

mean variance

$$\text{PDF: } f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \forall x \in \mathbb{R}$$

$$\int_{-\infty}^{\infty} f_X(x) \cdot dx = 1, \quad E(X) = \mu, \quad \text{Var}(X) = \sigma^2$$

- STANDARD NORMAL RANDOM VARIABLE ($N(0, 1)$)

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}}$$

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$$

$$\text{CDF: } \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

$$\text{Complimentary CDF} = 1 - \Phi(x) = P(X > x)$$

- Normality preserved under linear transformations:

$$Y = aX + b \Rightarrow Y \text{ is also gaussian.}$$

↪ gaussian

$$E(Y) = a \cdot E(X) + b$$

$$\text{Var}(Y) = a^2 \cdot \text{Var}(X).$$

$$Y \sim N(aE(X) + b, a^2 \cdot \text{Var}(X))$$

Q Suppose X is standard normal.

Find a and b such that $Y \sim N(\mu, \sigma^2)$.

$$\text{ans: } a = \sigma, \quad b = \mu$$

$$\text{CDF of } Y \text{ in terms of } X = \Phi\left(\frac{Y-\mu}{\sigma}\right)$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(ax + b \leq y)$$

$$= P\left(x \leq \frac{y-b}{a}\right) = \Phi\left(\frac{y-b}{a}\right)$$

* Central limit theorem

$$\frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

↪ always a bell-shaped curve

ANY random variable